$$T_{AA} = \ell_{AB}\, T_{BB}\, \ell_{BA}$$

$$= (\ell_{BA})^{-1} T_{BB}\, \ell_{BA}$$

$\Rightarrow$ T_{AA} and T_{BB} are similar matrix ☆

(A & B are similar $\Leftrightarrow \exists$ matrix S, $M = S^{-1} N S$)

$\Rightarrow$ T_{AA} & T_{BB} must have the same

eigenvalues, det, , $\chi_A(x)$, diagonalizability, ...

for any (∀) $T: V \to V$

$\Rightarrow$ (For instance): $\det(T) := \det(T_{AA})$

$= \det(T_{BB})$, ∀ A, B are bases of V

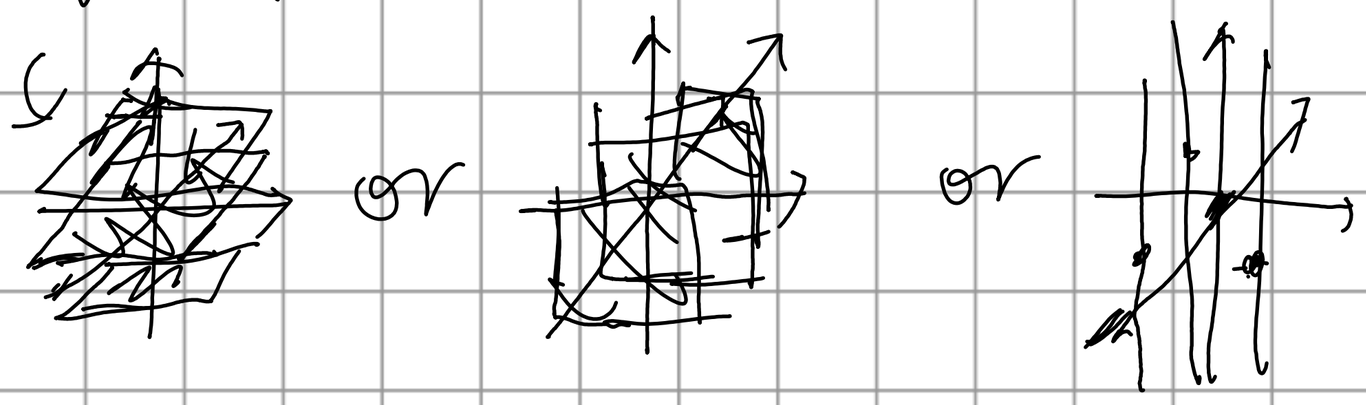
Some Constructions of Vector Spaces

Quotient Space V/W

Def.: collection of all cosets $\underline{v}+W$ (subspace)

Idea: Partition a vector space V into "subspaces"

E.g.: $V=\mathbb{R}^3$



Coset

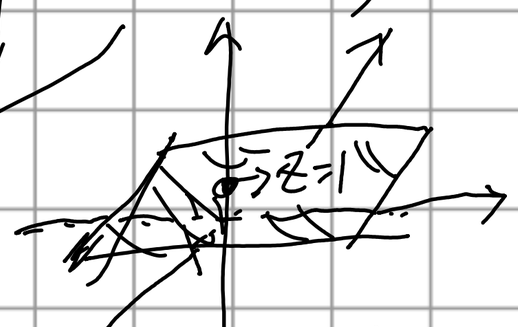
Def.: the (right) coset determined by $\underline{v}$ is the set $\underline{v}+W := \{\underline{v}+\underline{w} \mid \underline{w}\in W\}$ ($\forall \underline{v}\in V,\ W\leq V$)

E.g.: $V=\mathbb{R}^3$, $W=\text{Span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

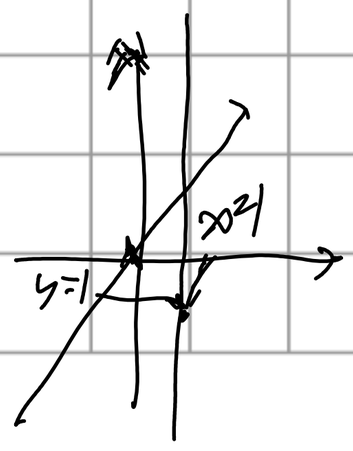
$$\begin{pmatrix}0\\0\\1\end{pmatrix}+W=\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}+\begin{pmatrix}a\\b\\0\end{pmatrix}\,\middle|\, a,b\in\mathbb{R}\right\}$$

$$\Downarrow$$

$$\begin{pmatrix}a\\b\\1\end{pmatrix}$$



$$W'=\text{Span}\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\} \Rightarrow \begin{pmatrix}1\\1\\0\end{pmatrix}+W'=\left\{\begin{pmatrix}1\\1\\a\end{pmatrix}\,\middle|\, a\in\mathbb{R}\right\} =$$



Remark 即使 $\underline{v} \neq \underline{v}'$, 也可能 $\underline{v}+W = \underline{v}'+W$

$\longrightarrow$ (same coset, different way to express)

Prop. (a): $\underline{v}+W = \underline{v}'+W \Leftrightarrow \underline{v}'-\underline{v} \in W$ ☆

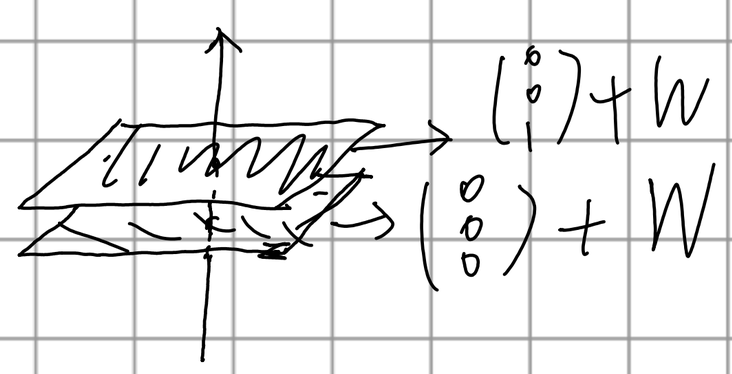
Prop. (b): If $\underline{v}+W \neq \underline{u}+W$

$\Downarrow$

then $(\underline{v}+W) \cap (\underline{u}+W) = \phi$

E.g.: $\begin{pmatrix}0\\0\\0\end{pmatrix}+W$ and $\begin{pmatrix}0\\1\\0\end{pmatrix}+W$

have no common vectors



Proof of Prop. (a):

($\Rightarrow$): By assumption, $\underline{v}'+W = \underline{v}+W$

$\Rightarrow \underline{v}'+\underline{0} = \underline{v}+\underline{w_0}$ ($\leftarrow \exists \underline{w_0} \in W$)

$\Rightarrow \underline{v}'-\underline{v} = \underline{w_0} \in W$?

($\Leftarrow$): Suppose that $\underline{v}'-\underline{v} = \underline{w_1} \in W$

$\forall \underline{w}, (\underline{v}'+\underline{w}) \in \underline{v}'+W$?

$\Rightarrow \underline{v}'+W \leq \underline{v}+W$

$\underline{v}'+\underline{w} = \underline{v}+\underline{w_1}+\underline{w} = \underline{v}+(\underline{w_1}+\underline{w})$, $(\underline{w_1}+\underline{w}) \in W$

$\in \underline{v}+W$

and (+) (Ex.) $\underline{v}+W \leq \underline{v}'+W$

?

Proof of Prop. (b):

Suppose that $\underline{x} \in (\underline{U}+\underline{W}) \cap (\underline{U}+\underline{W}$

(反证法)

Quotient Space

Ref.: Let $W \leq V$

The quotient space

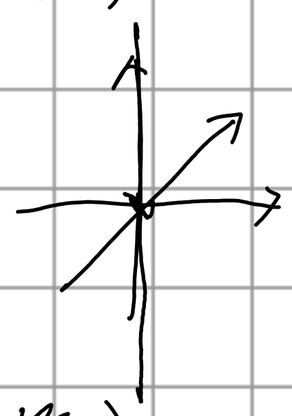
is the set $V/W := \{\underline{v} + W \mid \underline{v} \in V\}$

Remark 1 "element" in V/W

is 1 coset in V

E.g.: $V = \mathbb{R}^3$, $W = \mathrm{Span}\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$

$V/W = \left\{\begin{pmatrix}0\\0\\0\end{pmatrix} + W, \begin{pmatrix}1\\0\\0\end{pmatrix} + W, \begin{pmatrix}1\\1\\0\end{pmatrix} + W, \cdots\right\}$

$= \{$  ,  ,  , $\cdots\}$

① (V/W 的运算)

Def./Theorem:

① V/W is a v. space, with

scalar multiplication: $\alpha \cdot (\underline{v} + W) := \alpha\underline{v} + W$

"vector" addition: $(\underline{v_1} + W) + (\underline{v_2} + W) = (\underline{v_1} + \underline{v_2}) + W$

↳ (A "vector" in V/W $\leftrightarrow$ a coset in V

$\rightarrow$ (in vector addition)

coset + coset = coset)

(接上页)

Remark There's a big problem of well-definedness,

since $\exists v' \neq v$, $v' + W = v + W$

E.g.:

(exp. 不用, 实数相关 ~~(-)~~) $v_1 + W = v_1' + W$, $v_1 \neq v_1'$

$v_2 + W =$ ~~~

$\Rightarrow$ $(v_1 + W) + (v_2 + W)$ $(v_1' + W) +$ ~~~

$= (v_1 + v_2) + W$ $=$ ~~~

$\Rightarrow$ the addition is well-defined only when $(v_1 + v_2) + W = (v_1' + v_2') + W$

(otherwise: 1 sum, 2 outputs)

Proof (of $(v_1 + v_2) + W = (v_1' + v_2') + W$):

$v_1 + W = v_1' + W$, $v_1 \neq v_1'$

$v_2 + W =$ ~~~

① $\Rightarrow$ $\exists w_1 \in W$, $v_1 - v_1' = w_1$

$\exists w_2 \in W$, ~~~

$\Rightarrow (v_1 + v_2) - (v_1' + v_2') = (\sim - \sim) + (\sim - \sim)$

$= w_1 + w_2 \in W$

Prop. (a) $\Leftrightarrow$ ① $\Leftrightarrow$ ~~~ $=$ ~~~

(↓)

E.g.: (名詞)

Observation: (☆) (名詞)